

Professional Communication Guide

University of South Carolina (USC) & Midlands Technical College (MTC) Collaborative Semester Project

Why this matters: MTC students are the manufacturing partners for the USC design teams. The email record is the project record: it must allow a manufacturing partner to understand, quote, fabricate, inspect, and respond without avoidable ambiguity.

1. Scope and grading basis

This guide applies to every USC-to-MTC project communication and to every USC response to an MTC communication. The professional communication score is based on the official email record, including subject lines, recipients, attachments, response timing, and the technical completeness of information transmitted.

- The USC team receives one shared communication score by default. Individual adjustments within a team will only be made in exceptional situations.
- The team's primary point of contact is the accountable sender, not the sole contributor. All USC team members remain responsible for reviewing design information and supporting their team's primary point of contact in timely and accurate responses.

2. Required email protocol

- Official USC and MTC school email is the required channel for all project communication, including questions, decisions, action items, approvals, and design transmittals. Do not use personal email, text messages, social-media messaging, or unrecorded file-sharing channels.
- Each USC team will designate one student as the primary point of contact (POC) with their MTC team member. The POC leads cross-institution correspondence and keeps the full project team informed. The POC may only be changed due to extenuating circumstances (e.g. if a POC drops the course, a POC is unable to meet their duties due to a medical emergency, etc.).
- Every cross-institution email must copy all remaining USC and MTC team members, and the course instructors Gary Shannon. (shannong@midlandstech.edu) and Andrew Gross (andrewgross@sc.edu).
- Use a specific subject line: EMCH 308/MTC | Team [USC Team] - [MTC Team] | [Action or topic] | Rev [letter].

- Any discussion held in a meeting, phone call, or lab is informal only; it cannot replace official email communication. The USC POC must send the decisions, action items, owners, and due dates by official email within one business day. That email is the controlling record.

3. Response standard and issue management

USC students must send a complete reply within one business day of receiving an MTC email. A business day is Monday through Friday, excluding USC and MTC institutional holidays; messages received after 5:00 p.m. are due by the end of the next business day.

- A complete reply answers every question, acknowledges every action item, and identifies the responsible USC person when appropriate.
- If requested information cannot yet be provided, state what remains to be done and give a specific calendar date for the update. 'We will get back to you soon' is not a complete response.
- When an email contains three or more questions, decisions, or discussion items, number the items. Reply using the same numbers so that each item can be closed unambiguously.
- When a request is unclear, ask a concise clarification question rather than guessing. Confirm decisions that change geometry, material, cost, manufacturing method, datum scheme, or schedule.

Response test: Before sending, ask: Can the recipient tell which items are complete, which are pending, who owns each pending item, and exactly when the next update will arrive?

4. Manufacturing design-package requirements

The first formal design handoff to MTC must provide a manufacturing-ready package. Send both files in the same email and identify the revision in the email body and filenames.

Required item	Minimum expectation	Evidence in email record
3D model	Native .step file of the current part geometry.	Attachment with revision-controlled filename.
Manufacturing drawing	PDF with orthographic projections, complete GD&T for manufacturing, datum feature symbols, and tolerances tied to the datum scheme requested by MTC.	Attachment with matching revision-controlled filename.
MTC datum request	Use the datums requested by MTC on the first drawing shared. If the request is unclear, obtain written clarification before issuing the drawing.	Datum request and resulting drawing are traceable in the email thread.
Manufacturing context	Identify material/stock selection, revision, and any critical process, finish, inspection, or cost constraint that MTC needs to interpret the design.	Concise transmittal note or drawing note.

- A drawing is not complete merely because it resembles the project sketch. The drawing must define the geometry and tolerances needed to manufacture and inspect the part without relying on an informal interpretation of the 3D model.
- Do not overwrite an earlier version. Issue a new revision when design intent changes, state what changed, and identify whether MTC should stop work, continue work, or await confirmation.

5. Writing and document-control expectations

- Lead with the requested action or decision. State the project context in the first one or two sentences; use complete, professional sentences and a respectful tone.
- Name attachments descriptively, for example: USC03_MTC02_IBeam_RevA.step and USC03_MTC02_IBeam_RevA_Drawing.pdf.
- For a revised design, identify: (1) the revision, (2) what changed, (3) why it changed, (4) the anticipated effect on cost/manufacturing/testing, and (5) the action requested from MTC.
- Avoid vague words such as 'good,' 'small,' 'soon,' or 'as discussed' when a dimension, datum, date, name, or explicit decision is required.

6. Email templates

A. Initial manufacturing-package transmittal

Subject: EMCH 308/MTC | USC03 - MTC02 | Initial design package for manufacturing review | Rev A

Hello [MTC Team Member Name],

USC Team 03 is sending Rev A of our I-beam design for your manufacturing review. The attached STEP model and PDF drawing are the current controlling design package. The drawing uses datums A, B, and C as requested in your [date] email.

Please advise by [date] whether the stock selection, datum scheme, GD&T, and planned operations are feasible within the \$600 manufacturing-cost limit. If any feature needs modification, please identify the relevant drawing view and dimension/datum reference.

Thank you,
[USC POC Name] | USC Team 03

B. Numbered response to multiple items

Subject: EMCH 308/MTC | USC03 - MTC02 | Responses to manufacturing questions | Rev A

Hello [MTC Team Member Name],

Thank you for the questions below. Our responses are numbered to match your items:

1. Datum C: confirmed; Rev B will place the datum feature symbol on the lower flange surface.
2. Hole tolerance: we are checking the assembly requirement. We will send the final tolerance by Tuesday, March 17.
3. Stock option: we approve Option 3, subject to your confirmation that the remaining operations keep the estimate below \$600.

Please let us know if any item needs further clarification.

Thank you,
[USC POC Name] | USC Team 03

7. Communication grading method

Professional communication is a 15-point criterion in the project rubric. The team begins at 100% communication performance, and documented deductions below are applied to that score. The project points awarded equal 15 multiplied by the final communication-performance percentage. The score cannot be less than zero.

Assessment area	Share of communication score	What the instructor reviews
Channel, visibility, and ownership	20%	Official school accounts; designated representatives; complete CC list; traceable meeting summaries.
Responsiveness and follow-through	25%	Complete replies within one business day; clear update dates for open items; actions closed when promised.
Clarity and issue management	20%	Professional tone, specific subject lines, clear requested actions, and numbered treatment of three or more issues.
Technical transmission and document control	35%	STEP and PDF supplied; appropriate requested datums; complete GD&T; revision-controlled files and change notices.

8. Standard deductions for ineffective communication

The following schedule makes the grading consistent across teams. Deductions are percentage points from the communication-performance score and are cumulative when separate failures occur. The two 10% deductions below are required project rules.

Observed failure	Deduction	Application note
First PDF drawing does not use datums requested by MTC.	-10 percentage points	Applied once to the first drawing shared, even if corrected later.
Initial design-package email omits either the .step file or the PDF drawing.	-10 percentage points	Applied once per incomplete initial package; sending the missing file later does not remove the deduction.
Cross-institution communication uses a non-school account or omits required team/instructor CC recipients.	-5 percentage points per occurrence; maximum -10 per project phase	A project phase is initial handoff, revision cycle, or fabrication/testing closeout.
No complete reply within one business day.	-5 percentage points for each business day late; maximum -15 per required email response	A later reply is still expected; delay that blocks fabrication may warrant an additional -5 points.
Reply leaves a requested item unanswered and does not give a specific update date.	-5 percentage points	Applied once per email response.
Email with three or more issues does not number the issues, or reply does not follow the provided item numbers.	-3 percentage points	Applied when the omission makes tracking or closure unclear.
Unclear subject line, missing revision identifier, vague action request, unprofessional tone, or missing written summary of a project decision.	-2 to -5 percentage points	Instructor selects the smallest deduction consistent with the impact; repeated pattern may be treated as -5.

Individual accountability: The instructor may adjust a USC student's individual communication score when the record shows an assigned responsibility was not completed, a required review was ignored, or a representative repeatedly had to compensate for another student's lack of participation. The adjustment will be documented and normally will not exceed 5 project points without prior instructor consultation.

9. Pre-send checklist

- I used my official school email account and copied all USC and MTC team members plus both instructors.
- My subject line identifies the teams, action/topic, and revision when applicable.
- I answered every question or action item, and I gave a specific date for anything still pending.
- I numbered issues when there are three or more items to resolve.
- For a design handoff, I attached both the revision-controlled .step file and the PDF manufacturing drawing.
- The PDF drawing has complete GD&T and uses the datum scheme requested by MTC.
- A reader who was not present can understand the decision, owner, due date, and next action from this email alone.

10. Records to retain

Keep the official email thread intact through the end of the project. Instructors may request the thread or a PDF export to verify communication performance.

- Retain the MTC datum request, the first design-package transmittal, all later design revisions, and the email confirmation of the final approved configuration.
- Do not rely on a private notes file as evidence of a decision. If it affects fabrication, inspection, schedule, or cost, it must appear in the official email record.